

Evaluation of neutron spectra in the Linear Accelerator Facility of the Philippine General Hospital using an indium-based multi-shell neutron spectrometer (In-MuNS)

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Abstract

The neutron energy spectra of the 10 megavolt linear accelerator at the Philippine General Hospital (PGH) were evaluated using an indium-based multi-shell neutron spectrometer (In-MuNS). Measurements were conducted by employing an indium foil central detector within several high-density polyethylene spherical shells irradiated under fixed conditions. Indium activation foils utilize the neutron detection reaction $^{115}\text{In}(n,\gamma)^{116m}\text{In}$, emitting ^{116m}In gamma rays which were counted using a NaI(Tl) scintillation detector. Gamma spectrometry was then conducted to calculate the induced saturation activities. The Frascati Unfolding Interactive Tool (FRUIT) code, along with pre-validated In-MuNS response functions, were utilized to derive the neutron spectrum and ambient equivalent dose. The unfolding process revealed a total neutron fluence of $1.09 \times 10^5 \text{ cm}^{-2} \text{ Gy}^{-1}$ at the isocenter, and $4.29 \times 10^4 \text{ cm}^{-2} \text{ Gy}^{-1}$ at 70 cm away from the isocenter. The calculated ambient dose equivalent in air, $H^*(10)$, at the two locations are $17.75 \mu\text{Sv/Gy}$ and $5.60 \mu\text{Sv/Gy}$. The results can support local medical physicists in radiation protection against neutron contamination in LINAC facilities.

Keywords: medical LINAC, Bonner sphere

1 Introduction

Radiotherapy is one of the most common and effective methods to treat cancer using high-energy ionizing radiation to target tumors while minimizing the dose to surrounding healthy tissues. Advances in this field have been driven by linear accelerators (LINACs) with computer-assisted technology, which accelerate electrons to kinetic energies ranging from 4 to 25 MeV. However, using LINACs at potentials above 8 MV poses a radiation protection issue due to neutron contamination. Photonuclear reactions can occur when high-energy photons interact with high-Z materials of the LINAC components and in the treatment room, resulting in unwanted neutron exposure. Neutrons have a higher relative biological effectiveness (RBE) than electrons and photons, making even low neutron fluences significant for patient safety [1].

To address the carcinogenic risks of neutron contamination [2, 3], various studies have evaluated neutron fields around high-energy X-ray machines using the Bonner sphere spectrometer (BSS). BSS consists of a thermal neutron detector surrounded by polyethylene spheres of different diameters, each responding differently to neutron energy. Unfolding from BSS measurements determines the neutron energy spectrum. Neutron dosimetry studies have highlighted that photoneutron production varies depending on primary photon energy, LINAC design, and the treatment room [4]. Recognizing these differences, the neutron spectra in the treatment room of a 10 MV Varian Clinac CX LINAC model was measured using the indium-based multishell neutron spectrometer (In-MuNS), a modified BSS (mBSS) developed by the Philippine Nuclear Research Institute (PNRI). In the Philippines, this work is the second experimental neutron dosimetry evaluation at a medical LINAC facility and the first to evaluate a 10 MV LINAC. It follows the assessment of the 15 MV LINAC at the Lung Center of the Philippines by Fontanilla et al. in 2021, addressing the lack of current neutron spectra evaluations in local radiotherapy facilities.

2 Materials and Methods

In this study, the neutron spectra were measured from two measurement locations inside the treatment room of the 10 MV Varian Clinac CX linear accelerator located at the Department of Radiology of the Philippine General Hospital (PGH), Ermita, Manila, Philippines. As shown in Figure 1, Location 1 is the LINAC isocenter point, and Location 2 is a point 70 cm away from the isocenter towards the

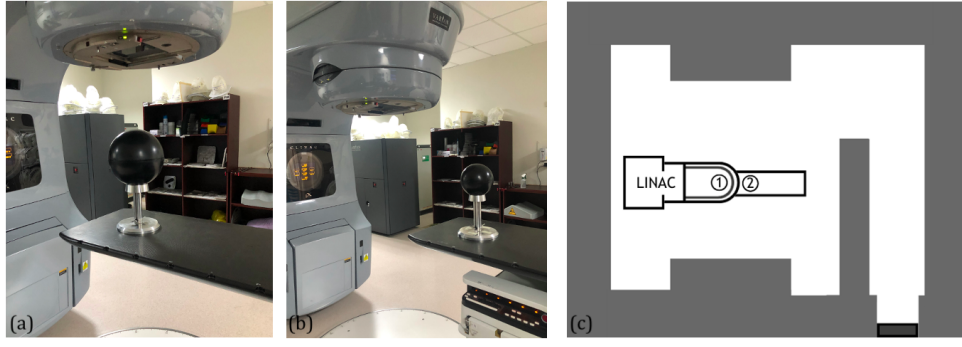


Figure 1: BSS experimental setups placed (a) at the isocenter point (location 1) and (b) at a point 70 cm away from the isocenter (location 2). (c) Layout of the LINAC treatment room and measurement locations 1 and 2.

“feet” direction of the couch. According to Fontanilla et al., Location 1 represents a point in space where direct photon irradiation occurs, while Location 2 represents a point outside the planning target volume (PTV) to account for the “out-of-field nontarget dose,” which is the dose deposited by stray or secondary radiation [5]. Measurements were conducted using the mBSS, which employs an indium foil as the passive central thermal neutron detector placed within high-density ($\rho = 0.939 \text{ g/cm}^3$) polyethylene (HDPE) hemispherical shells and central sphere that can be combined to form spherical moderators with 10 different diameters (5, 6.5, 8, 9.5, 11, 13, 16, 20, 25, and 30 cm). The activation foil technique utilized the detection reaction $^{115}\text{In}(n,\gamma)^{116m}\text{In}$ to measure the number of interactions induced by neutrons in the ^{115}In foils. Six foil-sphere configurations with sphere diameters, 8, 11, 13, 16, 20, and 25 cm were irradiated at each measurement location. Each configuration was irradiated with the LINAC operated at 10 MV with a $10 \times 10 \text{ cm}^2$ field size, 0° gantry angle, and a fixed photon dose of 2400 monitor units (MU) delivered at a photon dose rate of 400 MU/min. A waiting time of 15 minutes was allowed for foil retrieval. Gamma spectrometry was conducted for 180 minutes using a 3" x 3" NaI(Tl) scintillation detector (Alpha Spectra model 12I12/3) coupled to a 14-pin photomultiplier tube (*digiBASETM* compact module) controlled by the ORTEC Maestro-32 Advanced Multichannel Analyzer Software.

The induced saturation activity (A_i^{sat}) of ^{116m}In in $\text{Bq}\cdot\text{g}^{-1}$ was determined for measurements from the 1293.56 keV photopeak. The 1293.56 keV peak was selected for the unfolding as it has the highest photon emission probability and minimal self-absorption in the indium foil compared to other gamma peaks [6]. Unfolding was performed with the Frascati Unfolding Interactive Tool (FRUIT) developed by the INFN-Frascati National Laboratory [7], which derived the neutron spectra for each location through an iterative convergence algorithm utilizing Monte Carlo methods. The ISO-validated response matrix [8] for the In-MuNS facilitated the unfolding process, providing outputs such as neutron fluence, ambient dose equivalent $H^*(10)$, and fluence- and dose-averaged energies, along with associated uncertainties.

3 Results and Discussion

As shown in Figure 2, higher A_{sat} values were measured at location 1 because the neutron yield increases as the measurement point moves nearer to the LINAC head [5]. Compared to location 1, A_{sat} values decreased by about 58% in location 2, implying a decrease in total neutron fluence. Highest A_{sat} values were measured from the 13-cm sphere, followed by the 11-cm and 16-cm spheres in both locations.

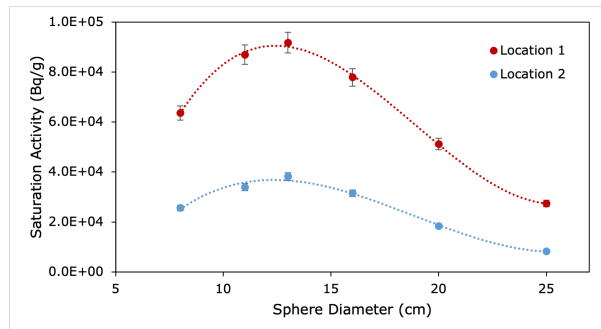


Figure 2: Induced saturation activities from 1293.56 keV gamma peak for each measurement location.

Figure 3 presents the neutron spectra in terms of unit lethargy derived from the calculated A_{sat} values at the 1293.56 keV gamma peak. The lethargy representation in a semi-logarithmic plot illustrates the relative distribution of the neutron components. The position of the fast neutron peaks above 0.1 MeV and of the thermal peaks at around 0.05 eV are in agreement with literature data [3, 5]. Intermediate neutrons are relatively minimal and no distinct peak was observed in this energy range.

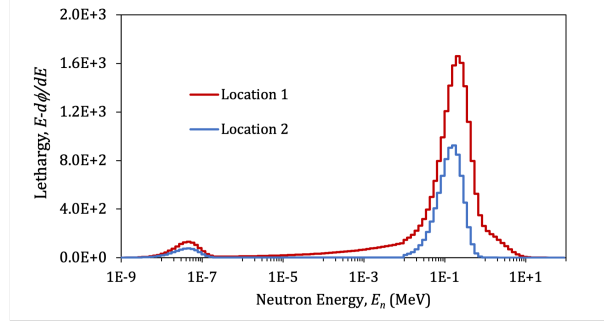


Figure 3: Lethargy spectra for the two locations normalized to total neutron fluence in a semi-logarithmic plot.

In a Monte Carlo analysis by McCall et al. [9] of the effects of a concrete room on the neutron spectrum, the neutron fluence components could be determined using the following empirical relationship:

$$\phi_{total} = \phi_{dir} + \phi_{sc} + \phi_{th} = \left(\frac{aQ}{4\pi d^2} \right) + \left(\frac{baQ}{S} \right) + \left(\frac{cQ}{S} \right) \quad (1)$$

where a is the transmission factor for neutrons penetrating the LINAC head shielding, d is the distance (cm) between the LINAC target to the point of measurement, S is the surface area (cm^2) of the treatment room, Q is the neutron source strength, and b and c are the coefficients depending on the materials of the walls.

Table 1: Neutron fluence fractions with respect to neutron energy at locations 1 and 2.

Classification (Energy Range)	Location 1	Location 2
Thermal Neutrons ($E < 0.625$ eV)	5.83%	5.59%
Epithermal Neutrons (0.625 eV $< E < 0.1$ MeV)	31.01%	31.22%
Fast Neutrons ($E > 0.1$ MeV)	63.16%	63.19%

The fluence fractions for each neutron energy interval in the unfolded neutron spectra are summarized in Table 1. In both spectra, the fast (direct) neutron component is dominant, and the thermal component is relatively minimal. Both locations are situated along the treatment couch and near to the LINAC target; hence, photoneutrons can freely exit the accelerator head and reach the detector without being thermalized by surrounding materials in the treatment room [5]. Since locations 1 and 2 are in proximity to each other, the fast neutron component only slightly varied. Still, the results agree with Eq. (1), which describes that the fast neutron component depends on the distance between the point of measurement and the target. The surface area of the treatment room also does not significantly affect the thermal and intermediate components at both locations, hence, only slight differences were observed. In essence, the fast neutron component indicates the major production of evaporation neutrons; the epithermal neutron component indicates the presence of evaporation neutrons that are slowed down due to interactions with the LINAC head and surrounding structures; while the thermal neutron component is mainly from neutrons detected due to room-return.

Table 2: Calculated integral quantities at locations 1 and 2.

Quantity	Location 1	Location 2
Total Fluence ($\text{cm}^{-2}\text{Gy}^{-1}$)	$(1.09 \pm 0.06) \times 10^5$	$(4.29 \pm 0.24) \times 10^4$
Ambient Dose Equivalent, $H^*(10)$ ($\mu\text{Sv Gy}^{-1}$)	17.75 ± 1.67	5.60 ± 0.62
Fluence-averaged Energy (MeV)	0.342	0.162
Dose-averaged Energy (MeV)	0.682	0.239

Table 2 shows the total neutron fluence and fluence averaged energy revealed in the unfolding process. The total neutron fluence in location 1 is a magnitude higher than in location 2. In radiotherapy treatment regimens, the total target absorbed dose delivered to the target tumor volume is typically 60 Gy. Considering neutron radiation, the ambient dose equivalent ($H^*(10)$) corresponds to an additional neutron dose of 1.07 mSv at the isocenter and 0.336 mSv at 70 cm away from the isocenter. Dose limits do not apply to medical patients, but the calculated neutron doses are close to the public dose limit of 1 mSv/yr established by the ICRP. This highlights the importance of monitoring potential neutron contamination in therapeutic photon doses during radiotherapy treatments. Neutrons, being uncharged particles, interact differently with tissue and can be more effective at causing biological damage compared to charged particles [1]. Therefore, the $H^*(10)$ values obtained from this work can assist local medical physicists in ensuring radiation protection.

4 Conclusions

The neutron spectra at two measurement points inside the treatment room of a 10 MV Varian Clinac CX linear accelerator were characterized using the locally developed indium-based multi-shell neutron spectrometer (In-MuNS). Integral quantities, in terms of neutron fluence, ambient dose equivalent, fluence- and dose-averaged energies, were obtained from the measurements. This work presents results on the neutron dose contribution during radiotherapy treatments, providing valuable data for enhancing treatment plan quality, reviewing and updating protocols to address neutron contamination, and improving facility optimization in radiotherapy.

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